End-Term Sample Question Paper - SET E

Course Code: CSE563 Course Title: APPLIED MATHEMATICS

Maximum Marks: 60 Duration: 2 Hours Set E - all six units, theory and numerical

Academic Task Type: End-Term Sample Date: _____ Registration Number: _____

Instructions: answer all questions. Part A carries 2 marks each, Part B 5 marks each and Part C 10 marks each; in Part C the marks of each sub-part are shown in brackets. Show all working for numerical parts — an unsupported final answer earns no marks. The unit, course outcome and Bloom level are printed against every question.

Part A - Very Short Answer Questions (5 x 2 = 10 Marks)

Q1. Define injective, surjective and bijective functions, and state which property a function must have for an inverse to exist. **(2 Marks)**

[Unit I · CO1 · L1 - Remember · Theory]

Q2. A covariance matrix has eigenvalues 12, 5, 2 and 1. Compute the percentage of variance explained by PC1, and by PC1 and PC2 together. **(2 Marks)**

[Unit II · CO2 · L2 - Understand · Numerical]

Q3. A regression model reports a correlation coefficient of r = 0.7. Compute R^2 and state precisely what it means. **(2 Marks)**

[Unit III · CO3 · L2 - Understand · Numerical]

Q4. State the max-flow min-cut theorem, and state what the capacity of any cut tells you about the maximum flow. **(2 Marks)**

[Unit IV · CO4 · L1 - Remember · Theory]

Q5. Arrange $n \log_2 n$, n^2 and 2^n in increasing order of growth by evaluating all three at n = 16. **(2 Marks)**

[Unit V · CO5 · L2 - Understand · Theory]

Part B - Short Answer Questions (4 x 5 = 20 Marks)

Q6. For an amount of 8 with denominations {1, 4, 5}, give the greedy coin-change solution and the true optimal solution, with the number of coins in each, and state what this shows about the greedy method. **(5 Marks)**

[Unit VI · CO6 · L3 - Apply · Numerical]

Q7. For $f(x, y) = 3x^2 + 2xy + y^2$: find the gradient, evaluate it at (1, 1), perform one gradient-descent step from (1, 1) with learning rate $\alpha = 0.1$, and verify that f decreased. **(5 Marks)**

[Unit I · CO1 · L3 - Apply · Numerical]

Q8. State the Central Limit Theorem, and explain when a t-test should be used in place of a z-test. **(5 Marks)**

[Unit III · CO3 · L2 - Understand · Theory]

Q9. Prove from the formal definition that $3n^2 + 2n + 5 = \Theta(n^2)$: find constants c_1, c_2 and a threshold n_0, and verify them at n = 1 and n = 5. **(5 Marks)**

[Unit V · CO5 · L4 - Analyse · Theory]

Part C - Long Answer Questions (3 x 10 = 30 Marks)

Q10. Matrices and linear systems. **(10 Marks)**

[Unit II · CO2 · L4 - Analyse · Theory + Numerical]

(a) State the condition for a square matrix to be invertible and give three equivalent characterisations, and state what it means for a system Ax = b when A is singular. *[5]*

(b) For A = [[2, 1], [5, 3]]: compute det A and A^{-1}, verify $AA^{-1} = I$, and solve $Ax = (4, 11)^T$. *[5]*

Q11. Graph traversal and topological sorting. **(10 Marks)**

[Unit IV · CO4 · L4 - Analyse · Theory + Numerical]

(a) Contrast breadth-first and depth-first search — the data structure each uses, the order each explores in, and one problem each is suited to — and state how a topological order is obtained from DFS. *[5]*

(b) Run DFS from A on the DAG with edges A→B, A→C, B→D, C→D, C→E, D→F, E→F, visiting alphabetically. Give the discovery and finish times and the topological ordering. *[5]*

Q12. Simulated annealing and the limits of optimisation. **(10 Marks)**

[Unit VI · CO6 · L5 - Evaluate · Theory + Numerical]

(a) Explain the exploration–exploitation trade-off, the role of temperature and the cooling schedule, and state what the No Free Lunch theorem asserts. *[5]*

(b) For $\Delta E = 8$, compute the Metropolis acceptance probability $\exp(-\Delta E/T)$ at $T = 50$, $T = 10$ and $T = 2$ to four decimal places, and state what the algorithm becomes as $T \to 0$. *[5]*

----- End of Question Paper -----